both confirm the expansion of the cosmos and map out its evolution. Translating redshifts to speeds, the pattern Hubble detected meant that the more distant a galaxy, the faster it's receding from us.

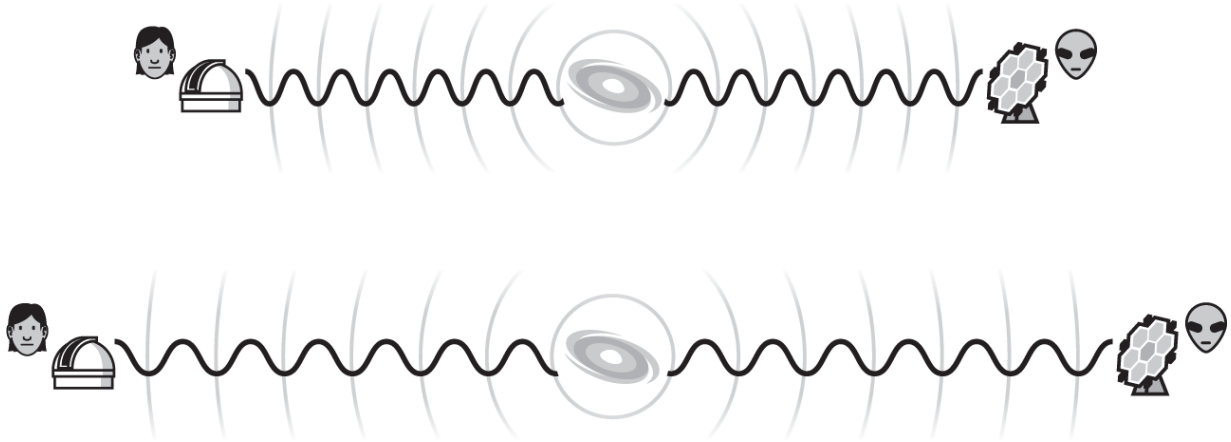


Figure 9: Cosmic expansion and redshift. As the universe expands, the light from distant galaxies is stretched out by cosmic expansion. This means that we will observe the light from a distant galaxy at a longer wavelength (redshifted) as the expansion of the universe proceeds. Because the expansion is happening everywhere, another observer watching a distant galaxy somewhere else in the cosmos will also see that galaxy's light redshifted.

Imagine stretching a slinky between your hands. (Just stretching, not bouncing. This is for science.) As you move your hands apart, each curl of the slinky moves only a finger's width away from the one next to it, but the two curls at opposite ends will end up a few feet apart in the same amount of time. If space is expanding uniformly in all directions, the same kind of relationship should hold, and that's exactly what Hubble's observations found. Mathematically, this gives us a conveniently simple rule of thumb: a galaxy's apparent speed is *directly proportional* to its distance. Meaning, first, more distant things are moving away faster. And second: there's a single number by which you can multiply any galaxy's distance to get its speed. While it was Hubble's data that ultimately proved the relationship and produced an estimate for that number, the proportionality was actually predicted theoretically a few years earlier by Belgian astronomer and priest Georges Lemaître. The relationship is accordingly known as the Hubble-Lemaître Law.[fn3] And the constant of proportionality (the number by which you multiply the distance) is called the Hubble Constant.

The crucial part for us here is the connection between redshift and distance. It means that we can look at a distant galaxy, measure the redshift,